PAPER_574 — Mayan 5-Cycle Cosmic Architecture & UQFF

Unified Quantum Field Framework (UQFF)

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Source: PAPER_574_Mayan_5Cycle_Cosmic_Architecture_Universal_Epoch_UQFF.md

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PAPER_574 — Mayan 5-Cycle Cosmic Architecture & UQFF

CP4 Class: companion to PAPER_573 (#161) **Session:** 154 **Cross-refs:** PAPER_573 (hub), PAPER_484 (Wolfram hypergraph), source116.cpp

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§1 Abstract

The five epochs of the Mayan Long Count Calendar (4 completed cycles + 5th epoch from 2012) correspond to five distinct phases of cosmic nuclear synthesis within the UQFF. Each epoch is characterised by a specific P_order range, n_cross complexity, and dominant nuclear mechanism. The 26th-order dimensional framework naturally produces 5 orbital shells which map onto the 5 calendar cycles. The post-2012 integration epoch corresponds to UQFF Epoch 5: superheavy element synthesis where buoyancy stabilisation enables otherwise inaccessible nuclear states.

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§2 Calendar ↔ Physics Mapping

$$\text{Epoch}_i \equiv \text{UQFF formation regime } i, \quad i = 1\ldots 5$$

Calendar epoch	Year range	UQFF regime	Element range	Dominant force
1 — Creation	~3114 BCE	$P > 0.999$	H, He, Li	U_m DPM pairs
2 — Growth	classical	$P > 0.1$	Be–Fe	T_j pyramid
3 — Conflict	dark ages	$P > 0.01$	Co–Xe	advanced coupling
4 — Transform	modern	$P > 0.001$	Cs–U	actinide resonance

Calendar epoch	Year range	UQFF regime	Element range	Dominant force
5 — Integration	2012+	$P > 0.0001$	Np–Og+	U_b stabilisation

§3 Probability Order by Epoch

$$P_{\text{order}}^{(i)} = \frac{e^{-S_i/\nu_{\max}}}{Z_i}, \quad S_i = k_B Z_{\text{mid},i}$$

Epoch 1 anchors: $Z_{\text{mid}}=2$ (He) → $P \approx 0.9998$ Epoch 5 anchors: $Z_{\text{mid}}=105$ (Db) → $P \approx 2.7\times10^{-4}$

§4 26D Dimensional Architecture ↔ 5 Epochs

The 5 epochs emerge from the 26D manifold's outer symmetry structure:

$$26 = 5\times5 + 1 \quad\text{(5 epochs of 5 shells + 1 integration threshold)}$$

Each shell group of 5 dimensions corresponds to one epoch. The 26th dimension is the integration singularity — the threshold of the 5th epoch (2012 in calendar terms).

§5 Epoch n_cross Complexity

3D-IPO crossing index by epoch:

$$n_{\text{cross}}^{(i)} \propto \frac{\ln Z_{\text{mid},i}}{P_{\text{order}}^{(i)}}$$

Epoch	Z_{mid}	n_{cross} (approx)	Complexity
1	2	1	Minimal
2	15	8	Moderate
3	40	14	High
4	72	21	Very high
5	105	26	Maximum

At Epoch 5, $n_{\text{cross}} = 26$ — the full 26-step hypergraph synthesis is required.

§6 Post-2012 Integration Epoch

The 2012 Long Count Calendar end-date coincides with the UQFF Epoch 5 activation threshold:

- Buoyancy term U_b becomes dominant over U_g for Z > 118
- SCm superconducting mode activated
- DPM antisymmetric pairs enable negative-mass nuclear corridors
- P_order drops below 0.18 for Z > 118 → instability unless buoyancy compensates

UQFF predicts that the Island of Stability at Z=119–126 (PAPER_577) is a direct consequence of buoyancy stabilisation active in Epoch 5, which the Mayan calendar predicted as the "Integration of polarities" final age.

Source: `grok_share_efc8a971378f.txt` — Session 154 *See also:* PAPER_573 (3D-IPO hub), PAPER_577 (island stability), source116.cpp (Wolfram hypergraph)